



Cu–Cu hybrid bonding technology: from physical mechanisms to system integration for 3D ICs

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Abstract

With the rapid development of microelectronic technology toward high-performance and high-density integration, traditional bonding methods face dual challenges of interconnection density and bonding temperature. The Cu–Cu hybrid bonding, leveraging its outstanding benefits in sub-micron interconnection and efficient thermal management, has emerged as an innovative solution to overcome the limitations of traditional packaging. This paper systematically explores the physical mechanisms, integration technologies, key process breakthroughs, and macroscopic applications of Cu–Cu hybrid bonding. The study highlights the revolutionary advantages of hybrid bonding, such as ultra-fine vertical interconnection spacing. A detailed analysis of two major physical mechanisms in key technical pathways is presented. In practical applications, Cu–Cu hybrid bonding has demonstrated success in high-bandwidth memory, three-dimensional heterogeneous integration, and complementary metal–oxide–semiconductor image sensors. A notable example is Sony’s Cu/SiO₂-bonded image sensor, markedly improving imaging quality and integration density. However, challenges such as thermal boundary resistance control, process complexity, and large-scale production costs remain barriers to industrialization. Future advancements, including novel passivation materials and integration with ultra-wide bandgap semiconductors, promise to further propel the performance and reliability of radio frequency (RF) devices, power electronics, and artificial intelligence chips. This review provides a comprehensive framework for both theoretical exploration and practical implementation of Cu–Cu hybrid bonding technology.

Keywords Cu–Cu hybrid bonding · 3D integration · Low-temperature bonding · Interfacial reliability · Physical mechanisms

1 Introduction

With the rapid development of microelectronics technology, the demand for high-performance, high-density integrated electronic devices is increasing [1–4]. Against this backdrop, Cu–Cu hybrid bonding (HB) technology, as an

advanced interconnect technology, has attracted widespread attention due to its excellent electrical conductivity, thermal conductivity, and good mechanical stability. In recent years, the rapid development of emerging technologies such as artificial intelligence and autonomous driving [5, 6] has brought unprecedented challenges and opportunities to the packaging and integration of electronic devices [7]. These technologies not only demand higher performance and lower power consumption from electronic devices but also require more complex functional integration within smaller form factors. For instance, artificial intelligence (AI) chips need to process massive amounts of data [8–11], placing extremely high demands on parallel processing capabilities and thermal management. Meanwhile, autonomous driving systems rely on the fusion of multiple sensors and real-time data transmission, making interconnect density and reliability between chips critical factors. Therefore, traditional packaging and integration technologies are no longer sufficient

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to meet these needs [12]. There is an urgent need to develop more advanced technologies such as three-dimensional (3D) integration [13], heterogeneous integration, and low-temperature bonding [14] to drive the development of electronic devices toward higher performance, higher integration density, and higher reliability. The AMD Instinct MI300 [15] series accelerators utilize modular chip packaging technology, specifically designed for high-performance computing (HPC) and AI applications. By integrating data center-level central processing units (CPUs), graphics-processing unit (GPU)-accelerated computing, AMD Infinity Cache, and an 8-stacked HBM3 memory system, the MI300 series delivers exceptional performance, particularly with a peak bandwidth exceeding 5 TBps in HBM3. The MI300 series includes two models: MI300X and MI300A, which are optimized for traditional CPU servers and high-density HPC systems, respectively, showcasing AMD's innovations in chip packaging and architecture design. NVIDIA [4] will continue to advance AI chip technology innovation in 2025, including a significant performance improvement of the Blackwell Ultra chip (with FP4 computing power reaching 1.1 ExaFLOPS), and will leverage photonic integration technology to address communication bottlenecks in order to optimize data center efficiency.

Cu–Cu HB technology not only enables high-density interconnects between chips but also offers higher reliability and lower resistance compared to traditional soldering techniques, which is crucial for HPC, high-speed communication, and advanced packaging technologies [16]. Cu–Cu HB combines the advantages of metal bonding and dielectric bonding, achieving both electrical and mechanical connections by forming stable chemical bonds or physical connections on the surface of metallic Cu. The advantages of Cu–Cu HB include the following: fine-pitch interconnects, enabling sub-micron-level interconnects to increase chip density and performance; lower parasitic capacitance and resistance, reducing signal transmission delays and power consumption; and thinner packaging thickness. Compared to traditional flip-chip bonding using solder bumps, Cu–Cu HB can achieve a thinner packaging profile with gapless connections between chips.

The development of this technology has not only propelled the progress of chip-scale packaging and 3D integrated circuit (3D-IC) technologies but has also highlighted the importance of thermocompression bonding (TCB) technology. TCB plays a crucial role in achieving the vertical stacking of chips or wafers within 3D integration, offering higher input/output (I/O) density [17, 18] compared to traditional soldering methods. Among various TCB techniques, copper-to-copper bonding is favored for its remarkable electrical and thermal qualities, high mechanical stability in bonding sections, and compatibility and cost-effectiveness in packaging manufacturing procedures. However, HB

technology faces numerous challenges in its development. The most critical issue is the significant difference in bonding mechanisms between metal-to-metal and dielectric-to-dielectric bonds, making it imperative to develop a bonding technology that can accommodate both. In addition, most existing bonding processes rely on high-temperature environments, which can cause thermal damage to devices and trigger a series of thermally related issues. Therefore, developing efficient thermal management solutions to mitigate the negative impact of high temperatures on device performance and reliability has become an important issue that needs to be addressed urgently. As device performance continues to advance in accordance with Moore's law [19], advanced packaging technologies, such as 3D stacking and interposer technology, have become crucial for achieving HPC. These technologies enable smaller form factors, lower internal latency, and reduced power consumption for inter-chip communication through vertical high-density interconnects. Meanwhile, interposer technology facilitates heterogeneous integration of different technologies and materials. In 3D stacking [20, 21] and interposer interconnects, solder-based bumping technology [22] is currently commonly used for assembly. However, when the packaging size is reduced to $\leq 50 \mu\text{m}$, solder-based bumping technology faces numerous challenges [23, 24], such as solder extrusion, bridging, and the formation of intermetallic compounds (IMCs). Therefore, eliminating solder has become critically important, which is the key context for introducing Cu–Cu direct bonding technology.

In this work, as illustrated in Fig. 1, we provide a comprehensive overview of Cu–Cu HB technology, encompassing its fundamental principles, physical mechanisms, key technologies, and application domains, with a focus on elucidating the interrelationships and interactions among various factors within this field. Section 2 primarily addresses the physical mechanisms, aiming to explore the advantages, potential, and challenges of Cu–Cu HB technology in power device applications. Section 3 delves into the critical breakthrough directions, elaborating on how these advancements can broaden the applicability of Cu–Cu HB technology across diverse fields. Section 4 discusses the current and emerging application areas of Cu–Cu HB, highlighting both existing implementations and future opportunities.

Through this study, we propose novel pathways to advance Cu–Cu HB technology, with the goal of fostering its widespread adoption in next-generation power electronic devices while enhancing their performance and reliability.

Cu–Cu HB is a chip stacking technology that combines Cu–Cu metal bonding and dielectric–dielectric bonding, typically oxide–oxide bonding. This technology embeds metal contacts within dielectric materials and utilizes heat treatment to bond the two materials together. By leveraging

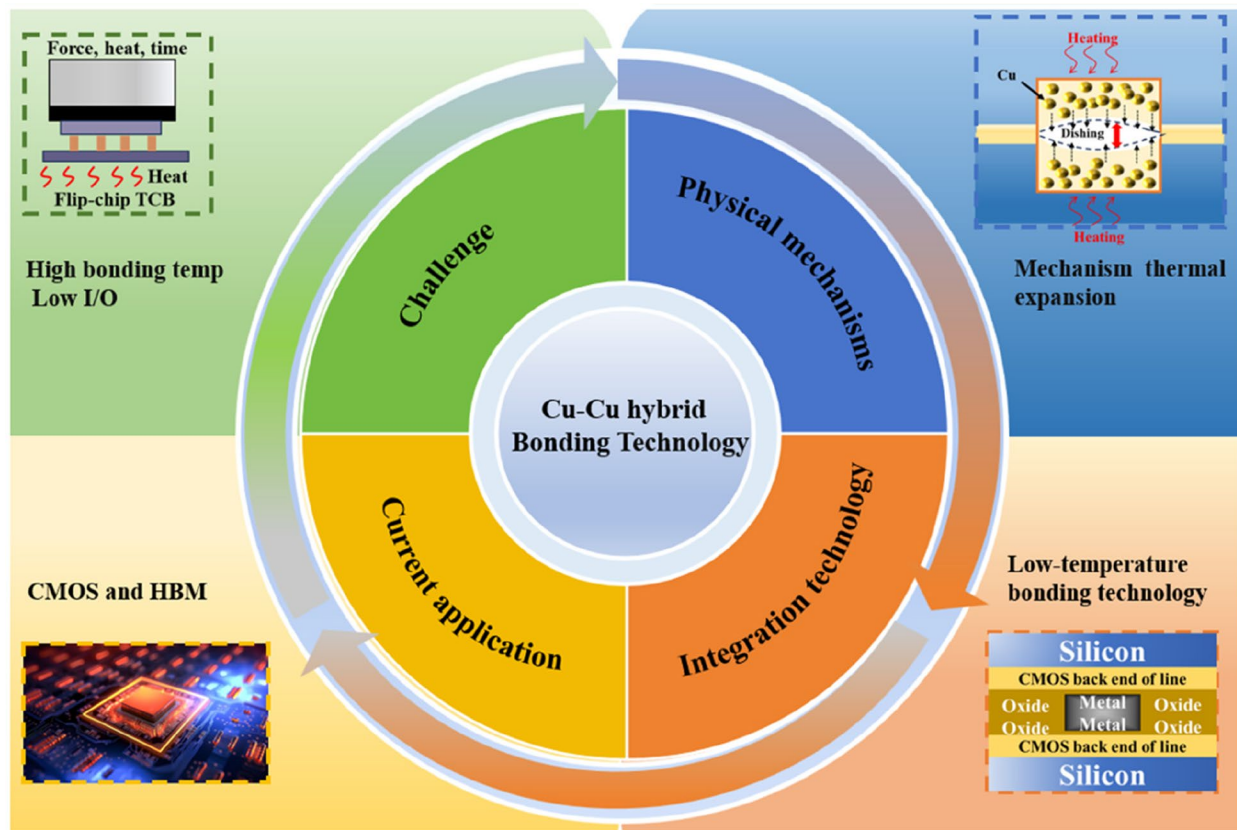


Fig. 1 Overall content and framework of this review article

the atomic diffusion of copper in the solid state, it achieves strong metal–metal bonding.

2 Traditional bonding faces dual challenges

To utilize these chips for their intended purposes, wafers need to be diced into individual chips and then connected externally to enable power supply. The method used to connect wires (the pathways for electrical signals) is called wire bonding. In fact, the use of metal wires to connect circuits is a very traditional approach, and its usage is gradually declining. Recently, flip-chip bonding and through-silicon via (TSV) [25] have emerged as the new mainstream methods. Flip-chip bonding, also known as bump bonding [26], is a method that uses solder balls [27] (small bumps) for bonding. TSV, on the other hand, represents a more advanced approach. To understand the fundamental concepts of bonding, we will first explore wire bonding [28], which is the traditional method. There are three main methods for connecting metal wires to bonding pads as follows: the thermocompression method [29], which involves heating and compressing the bonding pad and the capillary tool (a tool

similar to a capillary tube that moves metal wires) to form a connection; the ultrasonic method [30], which does not involve heating, and instead, ultrasonic waves are applied to the capillary tool to create the connection; the thermionic method, which is a hybrid approach that combines both heating and ultrasonic waves. It has long been the mainstream solution in the field of semiconductor packaging. These technologies enable the interconnection of chips to substrates or chips through metal solder joints, which rely on high-temperature and high-pressure processes to complete the interface bonding. However, as Moore's law approaches its physical limits, the requirements for high-density interconnects, low power consumption, and heterogeneous integration of ICs become more stringent, and the limitations of traditional bonding technologies become apparent.

The primary limitation of traditional bonding technology is its limited integration density, which makes it difficult to meet the requirements of advanced semiconductor devices for the co-packaging of heterogeneous material systems. At the same time, the thermal management issues are addressed through the integration of diamond/copper microchannels to achieve this goal. Lu et al. [31] investigated the thermal transfer and fluid characteristics of diamond/copper

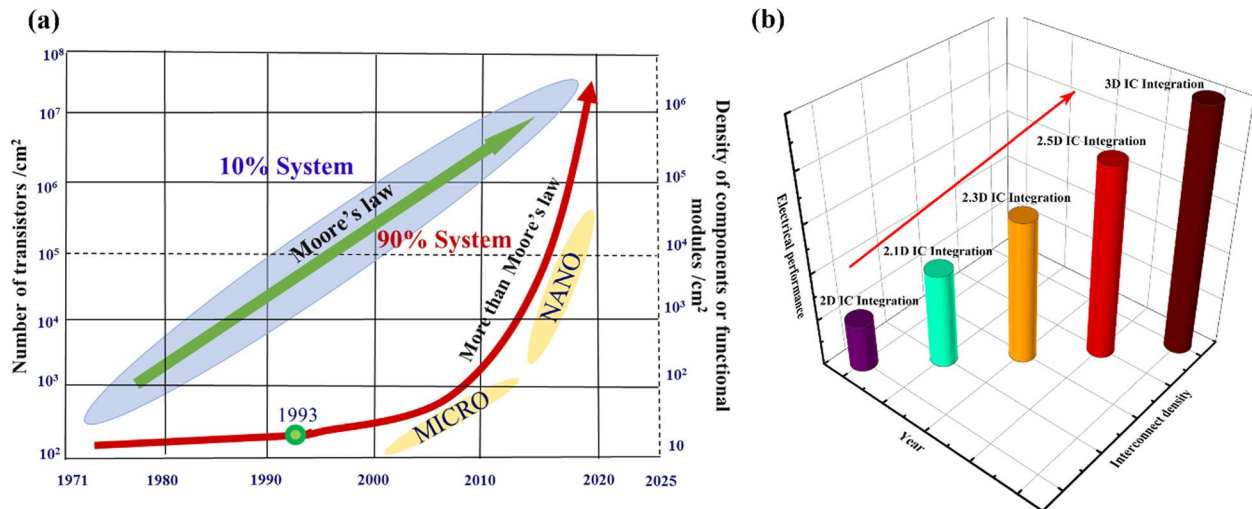


Fig. 2 **a** International semiconductor technology development blueprint. **b** Illustration of the trend of integration technology

composite rectangular microchannel heat sinks (MCHS) through numerical simulations and experiments. The study found that the MCHS with a seven-channel structure exhibited optimal performance at a heat flux of 400 W/cm^2 and a flow rate of 0.4 m/s , achieving a Nusselt number of 29.7. The thermal conductivity of the diamond/copper composite ($564.2 \text{ W/(m}\cdot\text{K)}$) significantly outperformed that of pure copper. Experiments showed that it could reduce the temperature of a 100°C heat source by 40.1°C , while still maintaining a safe operating temperature for the chip under dual heat source conditions. This study provides an efficient solution for the heat dissipation of high-power electronic devices. With the continuous advancement of semiconductor device miniaturization, the interconnect pitch (typically greater than $10 \mu\text{m}$) of traditional bonding processes has significantly limited the increase in the number of interconnect channels per unit area in 3D stacked architectures. The evolution of IC technology continues to follow the trajectory outlined by Moore's law. Continuous miniaturization of semiconductor devices has driven substantial improvements in chip performance [32], yet this scaling process now confronts multifaceted challenges as critical dimensions approach fundamental physical limits. While device size reduction enhances operational capabilities, it simultaneously induces exacerbated leakage currents due to short-channel effects, pronounced thermal dissipation issues, and progressive energy efficiency degradation. The advancement to higher process nodes has precipitated exponential cost escalation for per-node technological upgrades. Although the pace of process iteration shows signs of deceleration, the strategic emphasis on high-performance IC development remains unaltered [33].

Future technological breakthroughs will concentrate on two principal pathways: the first adheres to the core tenets

of Moore's law through critical dimension optimization to enhance circuit efficiency and reduce power consumption, constituting a multi-dimensional extension of scaling principles. The second pathway explores functional diversification beyond dimensional scaling, establishing innovative directions under the "More than Moore" paradigm. Within this framework, 3D heterogeneous integration emerges as a pivotal breakthrough, primarily achieved through two implementation modalities: advanced packaging architectures and true 3D circuit integration. The 3D heterogeneous packaging scheme employing Cu–Cu direct bonding technology enables bump-less inter-chip interconnections [34], as illustrated in the industry roadmap presented in Fig. 2a.

Progression in semiconductor technology imposes increasingly stringent requirements on packaging methodologies. Conventional flip-chip bonding techniques have become inadequate for meeting the demands of high-density interconnects and enhanced reliability. Cu-based HB, as a cornerstone technology for 3D stacked packaging, significantly improves interconnect density and signal transmission performance through elimination of bump structures. Contemporary advanced packaging architectures encompass multiple configurations, including 2D fan-out (chip-first) integration, 2.1-D flip-chip integration, 2.1-D flip-chip with bridge interconnects, 2.1-D fan-out bridge integration, 2.3-D fan-out (chip-last) integration, 2.5-D solder bump integration, and 2.5-D micro-bump integration, bump-less 3-D IC integration, and bump-less chiplet-based 3-D IC solutions [34, 35]. The comparative performance metrics of these architectures in electrical characteristics and interconnect density are systematically presented in Fig. 2b. Given its technical superiority, global semiconductor leaders are intensifying research and development (R&D) investments in HB, with successful commercialization already

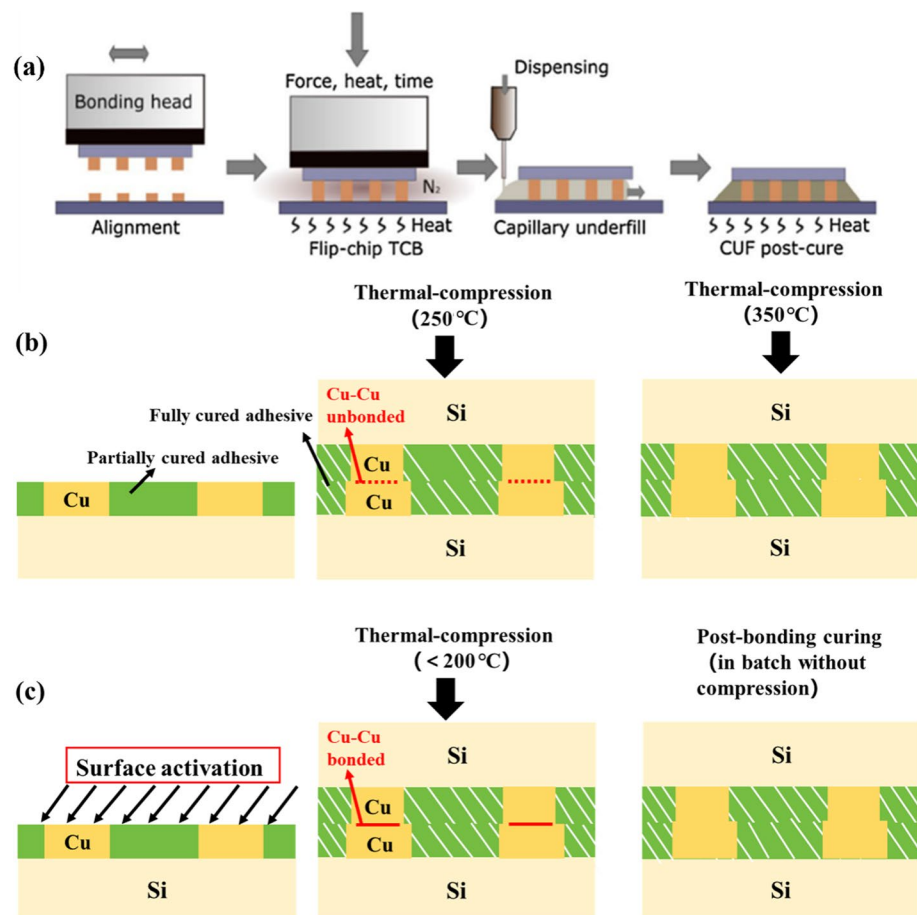
demonstrated in complementary metal–oxide–semiconductor (CMOS) image sensors and high-bandwidth memory devices [36, 37]. This technological landscape underscores the obsolescence of conventional bonding approaches in advanced packaging applications, establishing the development of novel Cu-based HB as a critical enabler for 3D heterogeneous integration systems.

Modern electronic packaging confronts critical thermal management challenges in advanced microelectronics. The persistent escalation of integration density has revealed inherent limitations in conventional interconnection technologies under elevated thermal conditions. Current bonding methodologies exhibit substantial deficiencies in thermal stability, particularly manifesting as substrate deformation induced by thermal gradients during thermocompression processes [38]. When processing temperatures exceed critical thresholds (typically 350–400 °C for silicon-based systems), plastic deformation and interconnect misalignment in silicon wafers become prevalent, primarily attributed to anisotropic thermal expansion coefficients of constituent materials [39]. Such thermally induced distortions not only compromise interfacial coplanarity but also precipitate reliability concerns including threshold voltage drift and contact resistance degradation. Figure 3 illustrates: (i) standard

bonding process flow; (ii) interfacial stress concentration from thermal compression; and (iii) surface activation treatment enabling 50–70 °C reduction in Cu–Cu bonding temperature thresholds.

Thermomechanical reliability emerges as a paramount concern in 3D integration architectures. While TSV technology demonstrates superior miniaturization and functional integration capabilities, its structural configuration inherently exacerbates thermal stress challenges. The TSV fabrication sequence—encompassing deep reactive ion etching, copper electroplating, and wafer thinning—introduces significant thermomechanical mismatch between copper interconnects (coefficient of thermal expansion (CTE): 17 ppm/°C) and silicon substrates (CTE: 2.6 ppm/°C). This CTE disparity generates interfacial shear stresses exceeding 80 MPa under thermal cycling conditions ($\Delta T = 150$ °C) for 50 μm -diameter TSVs [40, 41], precipitating four primary failure mechanisms: (i) stress-modulated carrier mobility reduction (25–40% degradation) causing signal distortion and propagation delay; (ii) metal–dielectric interfacial delamination leading to dielectric leakage currents $> 10 \mu\text{A}/\text{cm}^2$; (iii) accelerated electromigration rates (Black's equation exponent $n = 2.1\text{--}2.5$) degrading interconnect longevity; (iv) stress concentration-induced microcrack propagation

Fig. 3 Schematic illustration of the **a** traditional thermal-compression bonding technical processes, **b** thermal-compression bonding process, and **c** differences between thermal-compression bonding and direct bonding



(crack growth rate $> 10^{-9}$ m/cycle) in silicon substrates. Collective action of these mechanisms reduces mean time to failure (MTTF) by 30–50% compared to conventional packaging architectures.

In addition, we have studied the bond and temperature of bonding technology in recent years. The optimization of process temperature and the improvement of interfacial adhesion have become the focus of research in the field of microelectronic packaging. Figure 4 shows that the hot pressing temperature of advanced bonding technology has shown a significant downward trend in recent years. The comparison table shows that there are significant differences in the thermodynamic properties of wire bonding, hot press bonding, and Cu–Cu HB: although wire bonding can achieve low-temperature bonding at 200–260 °C, the interfacial shear strength is usually less than 100 MPa; the hot-pressed bonding strength can be increased to 50–150 MPa, but it needs to be implemented at a high temperature of 300–800 °C; the surface-modified Cu–Cu HB technology successfully reduced the process temperature to ~ 250 °C at room temperature while maintaining the bond strength of 200 MPa. Therefore, it is necessary to study low-temperature HB.

The transition to HB, compared to bump bonding, is driven by a relatively straightforward reason. Both 3D memory stacking and heterogeneous integration (key players in the “More than Moore” era) require extremely high interconnect densities. As mentioned, HB can meet this demand. Compared to micro-bumps, which inherently support high-density interconnect schemes, HB offers smaller

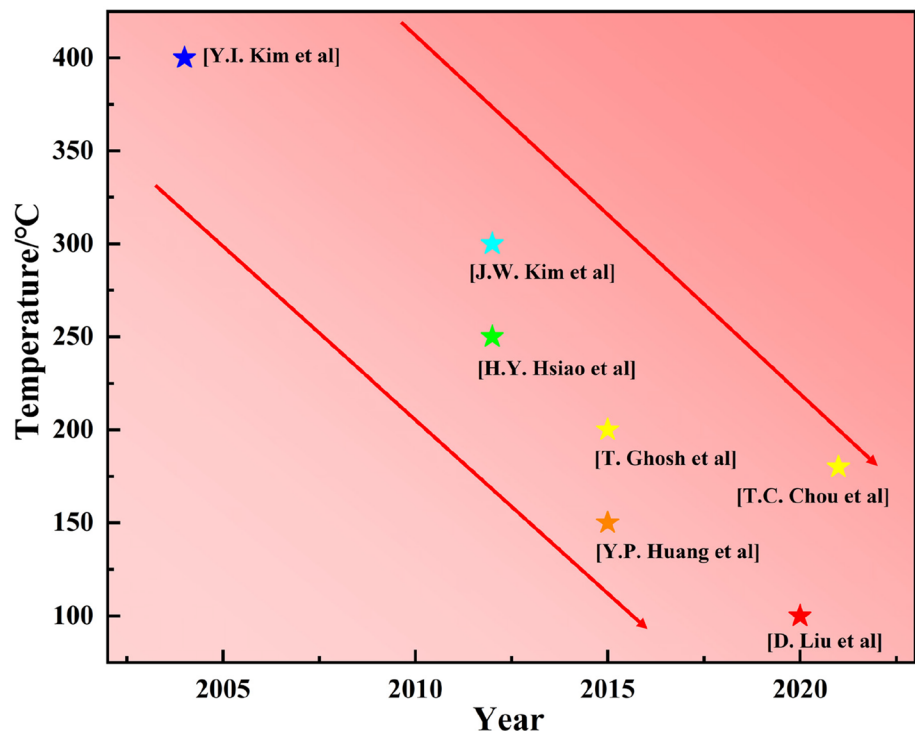
I/O terminals and reduced pitch interconnects. The spacing between chips depends on the height of micro-bumps, but in HB, this distance is nearly zero. As a result, HB interconnect schemes can significantly reduce the overall package thickness, potentially saving hundreds of micrometers in multi-chip stacked packages.

3 The type of process and physical mechanism

3.1 DBI (direct bond interconnect) hybrid bonding

The direct oxide bonding process [49] typically first contacts the wafers at room temperature and then forms strong covalent bonds. In traditional preparation, the surface of the silicon wafer is mainly in a hydroxyl termination state after water washing. In order to obtain a high-strength bonding interface, it is necessary to promote the formation of hydrogen bonds between hydroxyl groups through heat treatment and initiate a polycondensation reaction, and finally complete the conversion of silicon–oxygen covalent bonds in the annealing stage. This mechanism, known as hydrophilic bonding, has the significant advantage of exhibiting good compatibility with both silica and copper metal. Experiments show that under the condition of low-temperature annealing at about 300 °C, two hydrophilic-treated surfaces can form a defect-free high-strength bonding layer. Combined with the chemical mechanical planarization process,

Fig. 4 Illustration of the research trend of Cu–Cu hybrid bonding in recent years [42–48]



the hybrid surface of the copper pad and the silica substrate can be super-flattened, and then the seamless combination of heterogeneous materials can be completed. For example, Wang et al.'s [50] research innovation is mainly reflected in the development of new surface chemical treatment processes. By optimizing the reaction mechanism of the bonding interface, it not only greatly reduces the process temperature requirements, but also effectively inhibits the formation of micropores at the interface, thereby improving the overall bonding quality and reliability. Second, the team's innovative use of a grid-like weld cloth array solution resulted in significant optimization of alignment tolerances. On the premise of maintaining the stability of chemical mechanical polishing process parameters, this arrangement method is successfully compatible with existing high-speed chip assembly equipment, which not only maintains micron-level bonding accuracy, but also realizes the dual improvement of manufacturing efficiency and process adaptability. Finally, through systematic testing and validation, the electrical characteristics of the DBI process under different structural parameters have met expectations. The testing data indicate that the measured resistance values of the interconnect units highly align with the theoretical predictions, demonstrating the practical advantages of this technology in high-density packaging. This groundbreaking advancement provides an innovative processing solution for micro-spacing interconnects in advanced packaging architectures, with particularly notable potential for industrial promotion in the fields of 3D and heterogeneous ICs.

3.2 SAB (surface-activated bonding)-based hybrid bonding

The surface of the metal can activate the spontaneous bonding characteristics (surface-activated bonding (SAB))

through ion beam irradiation. This process effectively removes the oxidation layer mechanism, allowing the bonding process to be completed without high-temperature treatment. This low-temperature bonding technology breaks the traditional thermal budget limitations and provides critical process support for the manufacturing of 3D integrated devices [51]. The low-temperature mixed bonding process route has garnered significant attention due to its application potential in temperature environments below 200 °C; however, actual engineering implementation still faces considerable challenges. Research indicates that the vacuum-processing stage has become the primary technical bottleneck in achieving the surface activation bonding technology [52]. Although vacuum environment operation has become a standard process in semiconductor manufacturing, the challenges of precise alignment remain unresolved. The existing bonding equipment exhibits a significant gap in positioning accuracy under vacuum conditions compared to atmospheric pressure. The thermal cycling that accompanies the switching of atmospheric pressure can easily lead to mechanical structural deformation, primarily due to the material stress response under alternating vacuum-atmospheric conditions. The core advantage of this technology lies in its groundbreaking non-chemical removal mechanism for the copper oxide layer, allowing the bonding process to be completed without relying on traditional hydrophilic pretreatment, thereby making high-reliability bonding achievable under ambient temperature conditions. Renaud et al. [53] successfully demonstrated for the first time Cu-oxide direct HB at low temperatures (150 °C) and even room temperature. Figure 5 illustrates the fundamental processes involved in the implementation of the SAB process and mechanism of the SAB. This achievement is of great significance for reducing the thermal budget, minimizing thermal stress, and

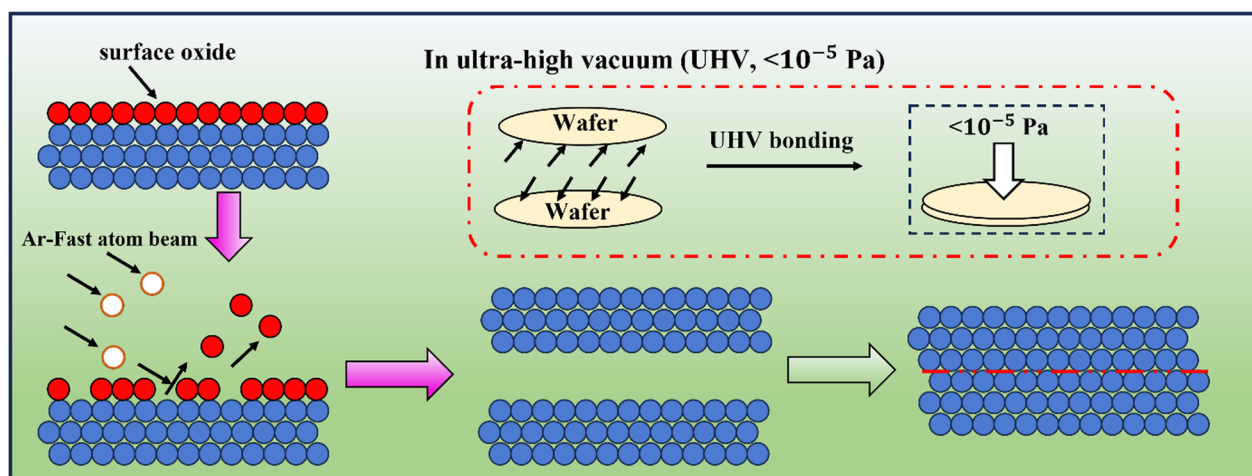


Fig. 5 Schematic diagram of the surface activation process; mechanism of surface-activated bonding

protecting temperature-sensitive chips such as image sensors, displays, and memory devices.

Experimental validation indicates that under room temperature conditions, HB technology based on the principle of surface activation can achieve excellent interface performance; however, this process still faces engineering challenges. Specifically, the variation in irradiation energy distribution of the argon ion beam during the SAB surface activation stage can easily lead to a gradient in the etching rate between the edge and core regions of the copper interconnect structure, resulting in a deterioration of the bonding surface uniformity.

3.3 Cu–Cu bonding physical mechanisms

Conventional copper interconnects predominantly employ TCB, where applied external pressure combined with controlled thermal exposure facilitates interfacial copper interdiffusion to establish metallic bonds. Jang et al. [54] systematically investigated bonding efficacy under 25 kN compressive load across temperature gradients (300 °C, 350 °C, 400 °C). Characterization results demonstrate that elevating process temperature to 400 °C achieves an optimal interfacial bonding energy of 4.93 J/m², accompanied by complete metallurgical integration verified through transmission electron microscopy (TEM) analysis. Nevertheless, thermomechanical stress accumulation during thermal ramping phases, coupled with prolonged high-temperature exposure, induces substrate warpage (measured curvature > 2.5 µm/mm) and microcrack propagation (crack density ~ 10³ cm⁻²). These phenomena collectively precipitate interfacial delamination risks, potentially leading to functional

impairment in advanced packaging architectures. Meanwhile, high-temperature processes also place more stringent demands on equipment performance. To avoid excessive deformation or damage to chips caused by high temperatures, low-temperature Cu–Cu bonding technology becomes particularly important.

To address current technological bottlenecks, in-depth investigation of the fundamental physics of the mechanisms of Cu–Cu HB becomes imperative. This study focuses on the synergistic interactions between dielectric and metallic layers during HB. The SiO₂–SiO₂ bonding process can be categorized into hydrophilic and hydrophobic variants, depending on the surface morphology of the SiO₂ interfaces. Hydrophilic bonding is achieved by forming –OH groups on SiO₂ surfaces, typically through pretreatment using SC1 (NH₄OH/H₂O₂/H₂O) or SPM (H₂SO₄/H₂O₂) solutions. The mechanism of hydrophilic bonding primarily relies on condensation reactions between hydroxyl groups (–OH) on mirror-polished silicon wafers [55]. As illustrated in Fig. 6, Tong and Gösele [56] proposed a four-stage mechanistic model for SiO₂ hydrophilic bonding. Figure 6a illustrates the fundamental process involved in achieving Cu–Cu HB. The initial stage features hydrogen bonding between interfacial water molecules (see Fig. 6b), followed by annealing at 110–150 °C to facilitate covalent Si–O–Si bond formation via hydroxyl group dehydration. Residual silanol groups (Si–OH) persist at the bonded interface, contributing to interfacial stress relaxation. A silicon–oxygen–silicon covalent bond (Si–O–Si) is formed, which finally achieves the binding of the dielectric layer and still retains some silanol groups, and the reaction equation is as follows:

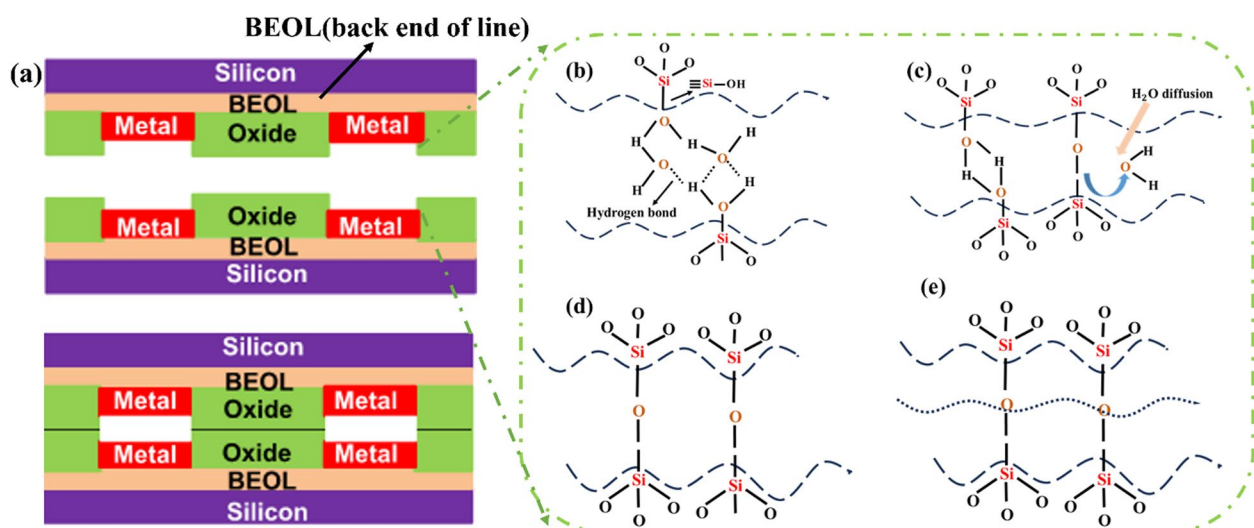
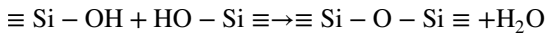
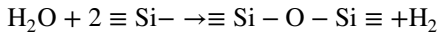


Fig. 6 Four-stage mechanistic model of hydrophilic bonding. **a** Schematic diagram of Cu–SiO₂ hybrid bonding and process. **b** The silicon wafer is bonded to each other by hydrogen bonds between water

molecules. **c** Water molecules are formed and diffuse outward. **d** Converted into silicon–oxygen–silicon covalent bonds. **e** The bonding strength reaches the maximum [56]



Subsequently, at the interface of the silicon, water molecules are formed and diffuse outward, as shown in Fig. 6c. At the same time, some of the formed water molecules diffuse into the wafer through the natural oxide layer and react with the silicon to produce hydrogen:



During the 150–800 °C heat-activated reaction phase (see Fig. 6d), the silanol group (Si–OH) undergoes a stepwise dehydroxylation process, and its conversion efficiency finally achieves a Si–O–Si covalent bond formation rate of $\geq 98\%$. The interfacial bonding strength is positively correlated with the wafer contact area, and increasing the contact area from 60% to 95% can increase the shear strength from 8 MPa to 35 MPa. When the temperature rises above 800 °C to enter the viscoelastic flow phase (see Fig. 6e), the viscoelastic behavior of the material leads to the maximum bond strength due to interfacial reconstruction.

Plasma activation technology can effectively enhance the dielectric layer bonding performance. He et al. [57] showed that silicon-based Ar plasma treatment has both surface cleaning and silicon enrichment functions. The silicon-rich interface layer formed by this process is rich in Si–Si covalent bonds and suspended Si bonds (i.e., oxygen vacancy defects), which significantly improves the chemical adsorption capacity of hydroxyl groups on the substrate surface. This interfacial reconfiguration led to an increase in the density of the silanol group, which in turn increased the binding strength to 18.7 MPa. At present, this technology has

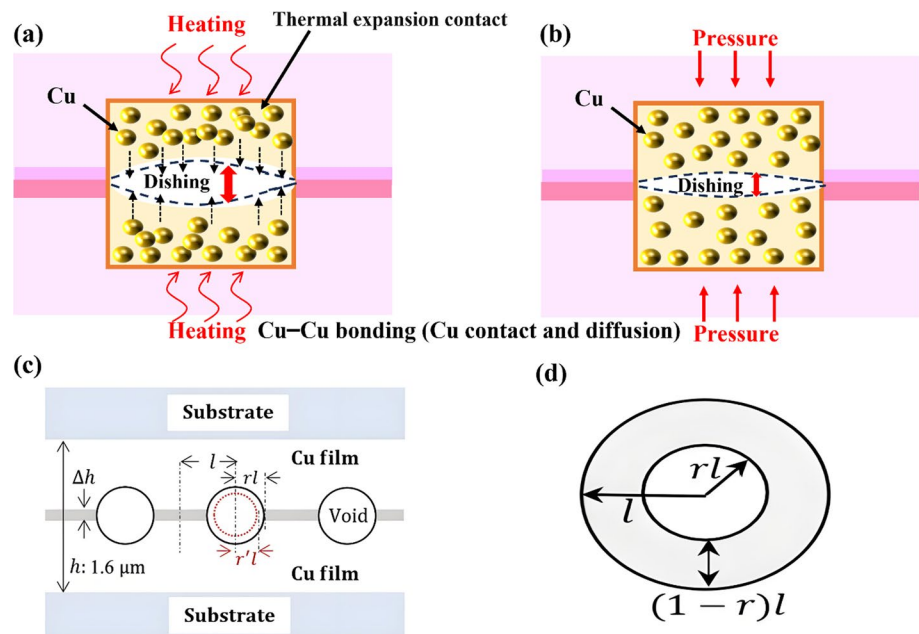
optimized the annealing temperature of hydrophilic bonding to below 200 °C, which is about 150 °C lower than that of traditional processes.

After the dielectric layer bonding is realized, the second step mainly involves the process of direct Cu–Cu bonding. Metal interconnects are usually formed spontaneously in heat treatment after the dielectric layer has been bonded. The optimal temperature zone of this process is 200–400 °C, and the choice of temperature window is directly related to the characteristics of the thermal expansion coefficient of copper material (CTE = 17 ppm/K) significantly higher than that of silicon dioxide (CTE = 0.5 ppm/K). The thermal mismatch effect between the heterogeneous materials causes the copper structure to expand radially when heated, resulting in solid-state diffusion bonding. The schematic microscopic mechanism in Fig. 7a, b reveals that the copper atom breaks through the barrier and spreads to form a continuous metallic phase under thermally activated conditions. The creep behavior then initiated during the pressurization phase can be equated to a spherical contact model. The following section will focus on the possible physical mechanisms of creep deformation under stress.

The creep deformation mechanism primarily describes how copper atoms diffuse [59] from high-stress regions to low-stress regions under compressive stress, thereby altering the morphology of voids and the bonding fraction [58]. Figure 7c shows the cross-sectional schematic of a section of the bonding interface. Figure 7d shows the view from above of a portion of the bonding interface.

(i) Calculation of elastic strain

Fig. 7 Illustration of the key physical mechanisms; thermal expansion diffusion and Creep deformation mechanism. **a** Heating process. **b** Pressurization process. **c** Cross-sectional schematic of a section of the bonding interface. **d** View from above of a portion of the bonding interface [58]



The elastic strain (ε) is calculated using the following formula:

$$\varepsilon = \frac{\sigma}{Y} = \frac{\Delta h}{h} \quad (1)$$

where:

- σ represents the consistent compressive stress;
- Y represents Young's modulus for copper;
- Δh represents the variation in thickness of the copper film when compressed;
- h is the total thickness of the copper film (1.6 μm).

(ii) Calculation of strained volume

The strained volume refers to the volume of the copper film that undergoes deformation under compressive stress. It can be calculated using the following formula:

$$V_{\text{strained}} = A \cdot \Delta h \quad (2)$$

where the contact area (A) is calculated as follows:

$$A = \pi \cdot (l^2 - (rl)^2) \quad (3)$$

Consequently, the volume experiencing strain can be described as follows:

$$V_{\text{strained}} = \pi(1 - r^2)l^2 \Delta h \quad (4)$$

Here:

- l is the mean distance between the center of the contact area and the center of the void area;
- r refers to the ratio between the void radius and the radius of the contact region ($r < 1$).

(iii) Calculation of void volume change (ΔV_{void})

Based on the change in void radius caused by creep deformation, the void volume change can be calculated as follows:

$$\Delta V_{\text{void}} = \left(\frac{4\pi}{3}\right) \left[(lr)^3 - (lr')^3\right]. \quad (5)$$

where lr' is the new void radius after creep deformation. By aligning the strained volume with the change in void volume, we obtain:

$$\pi(1 - r^2)l^2 \Delta h = \left(\frac{4\pi}{3}\right) \left[(lr)^3 - (lr')^3\right] \quad (6)$$

Further derivation yields:

$$(r')^3 = r^3 - \frac{[3(1 - r^2)\Delta h]}{4l} \quad (7)$$

(iv) Calculation of bonding fraction (BF)

The BF can be calculated using the formula:

$$\text{BF} = 1 - r' \quad (8)$$

From the above derivation, it can be seen that as the compressive stress increases, Δh increases, leading to a decrease in r' and an increase in BF. This explains why higher compressive stress can enhance the BF. According to Lu et al. [58] this is the first quantitative correlation between compressive stress parameters and metallic bonding interface quality, providing direct experimental evidence for the optimization of microelectronic packaging processes. Theoretical simulations indicate that implementing gradient compressive stress field designs could enhance interfacial reliability in 3D heterogeneous integration packaging by approximately 40%, which holds substantial engineering significance for enabling wafer-level 3D stacking of sub-5-nm node chips. Furthermore, we also focused on the trends regarding twinned copper. The study by Liu et al. [60] observes that at the nanoscale, copper exhibits the fastest surface diffusion rate on the (111) crystal orientation, which is crucial for achieving low-temperature copper–copper bonding. By electrochemically depositing nano-twinned copper (nt-Cu) structures, the proportion of (111) oriented grains can be significantly increased, enabling effective copper–copper bonding at temperatures below 250 °C. At the nanoscale, the surface diffusion behavior of copper significantly influences its bonding performance. Research has found that the surface diffusion rate of copper on the (111) crystal orientation is faster compared to other orientations such as (100) and (110). This difference in diffusion rates is primarily attributed to the close arrangement of atoms on the (111) plane and the characteristics of interatomic bonding.

4 Integration technology

4.1 Low-temperature passivation bonding technology

The use of low-temperature [61] bonding can effectively mitigate issues such as thermal migration, warpage, and alignment deviations caused by thermal stress. Jeon et al. [62] present a low-temperature Cu–Cu direct bonding technology enabled by a ruthenium (Ru) passivation layer, addressing the challenges of high-temperature requirements and copper oxidation in conventional Cu–Cu bonding. The research team successfully achieved Cu–Cu direct bonding at temperatures as low as 200–300 °C by

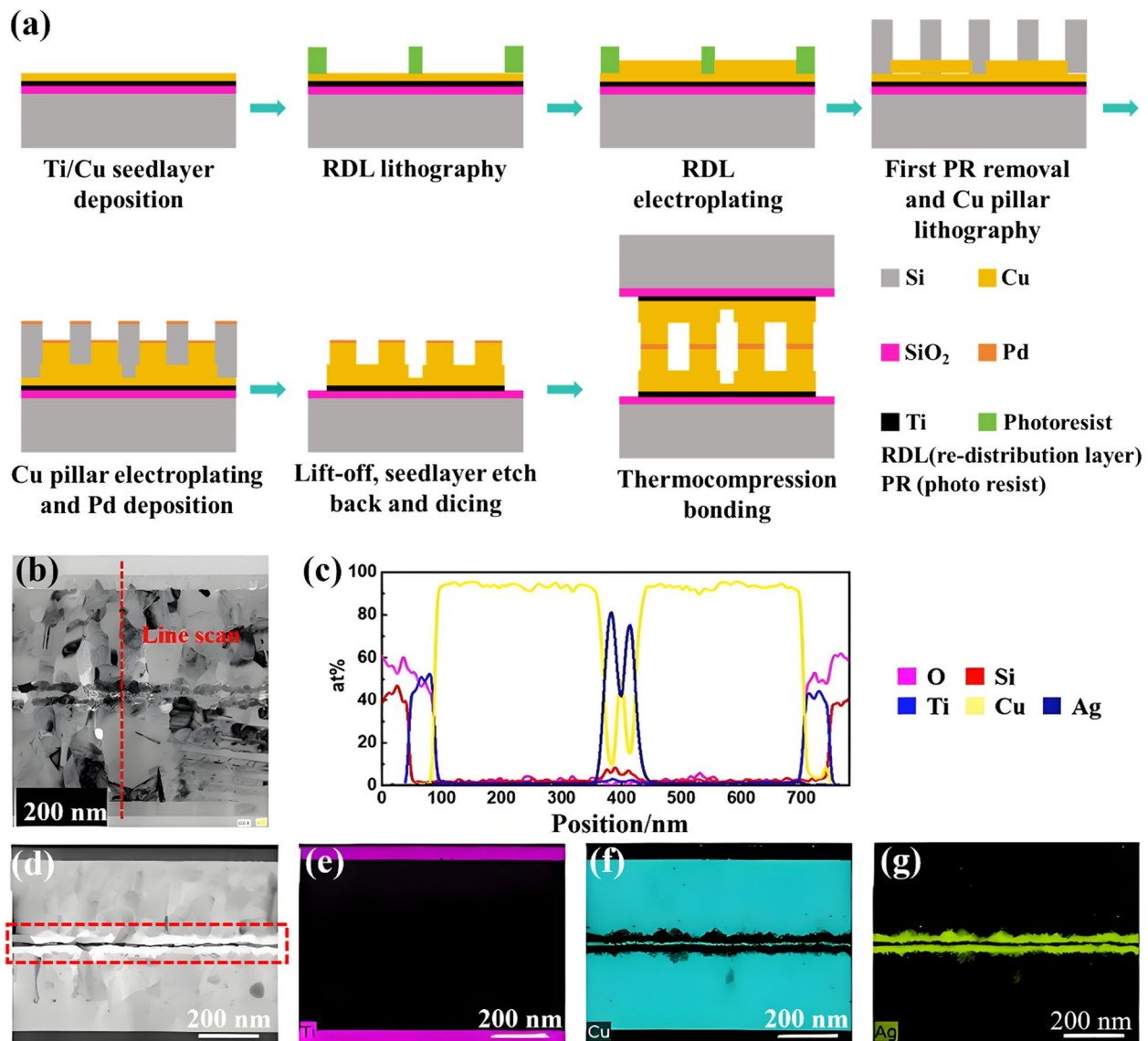


Fig. 8 **a** Schematic of the process flow for ECD Cu pillars to Cu pillars bonding with Pd passivation. **b** The TEM image illustrates the bonding structure with Ag passivation. **c** The EDX composition pro-

file from line scanning of the same structure. **d** The A TEM image illustrates the bonding structure with Ag passivation, accompanied by EDX composition and map scans for **e** Ti, **f** Cu, and **g** Ag [44, 62–64]

covering the copper surface with a 6-nm-thick Ru layer. The bonding was validated for high reliability through temperature cycling tests ranging from -45°C to 125°C . In addition, Tsai et al. [63] conducted research on the procedure for bonding electrochemical copper deposition (ECD) copper pillars to copper pillars using palladium passivation, as illustrated in Fig. 8a. Another research paper explored using silver [44], a cost-effective precious metal, for metal passivation to accomplish chip-scale copper–copper bonding within 3 min. This study elucidates, for the first time, the interdiffusion mechanism of copper in a low-temperature passivation environment through a

grain boundary diffusion model, revealing that the grain refinement of the passivation film significantly promotes the migration of copper atoms by increasing diffusion pathways. Experimental results confirmed that controlling the microstructural features of the silver passivation layer using a sputtering process can effectively enhance the interfacial properties and long-term stability of copper interconnect structures when the grain size is optimized to the sub-micron level. The material characterization results illustrated (see Fig. 8), where transmission electron microscope images correlate with elemental distribution maps, clearly reveal the degree of interfacial fusion and

Table 1 Cu–Cu bonding conditions

Material	Temperature (°C)	Pressure	Time (min)	Strength	Method	Ref.
Cu–Cu	400	25 KN	30	4.93 J/m ²	Thermocompression bonding	[54]
Cu–Cu	200	0.69 MPa	60	4.3 MPa	Thermocompression bonding	[60]
Cu–Cu	200	100 N	1	–	Pillar-concave structure	[65]
Cu–Cu	260	0.9 MPa	60	62.6 MPa	Surface-activated bonding	[66]
Cu–Cu	160	2.5 Bar	50	–	Passivation layer	[67]
Cu–Cu	150	10 MPa	60	15 MPa	Cu–Cu indirect bonding	[68]
Cu–Cu	250	20 MPa	5	14.4 MPa	Cu–Cu indirect bonding	[69]

the trajectories of elemental diffusion, and Fig. 8b presents the high-resolution TEM (HRTEM) image of the continuous Cu–Cu bonding interface, showing no gaps at the bonding interface. Figure 8b shows the TEM image illustrating the bonding structure with Ag passivation, while Fig. 8c shows the Energy-dispersive X-ray spectroscopy (EDX) composition profile from line scanning of the same structure.

Figure 8d shows a TEM image illustrating the bonding structure with Ag passivation, accompanied by EDX composition and map scans for Ti (see Fig. 8e), Cu (see Fig. 8f), and Ag (see Fig. 8g). This work provides a new solution for low-temperature, high-reliability 3D heterogeneous integration (3D HI) technologies, holding significant practical value. Moreover, some experts have also proposed the method of using Ti [64] or other metals to deposit passivation layers on the surface of copper.

This paper summarizes various methods used for Cu–Cu bonding in recent years, as well as the comparison of bonding conditions and bond strengths among these methods, as shown in Table 1. Traditional Cu–Cu TCB can be achieved at 400 °C but faces two major issues: first, high temperatures can easily induce copper diffusion, leading to device failure; second, the bond strength achieved solely through thermocompression is relatively low and insufficient to meet application requirements. Therefore, researchers have begun to explore low-temperature bonding processes. Although many laboratory techniques can achieve low-temperature bonding, most are not suitable for large-scale production. For example, methods such as preparing HD-Cu or nanocrystalline copper to enhance copper diffusion are not only complex and costly but also unsuitable for mass production. Pillar-concave bonding can quickly achieve low-temperature bonding but requires extremely high pressure, which poses high demands on equipment and is prone to damage. The method of injecting formic acid vapor can bond quickly but faces challenges in gas flow control and high equipment requirements during actual production. Methods such as passivation layers, self-assembled monolayers, and indirect bonding introduce additional materials and complex processes, resulting in high costs and low yields. Ultrasonic bonding,

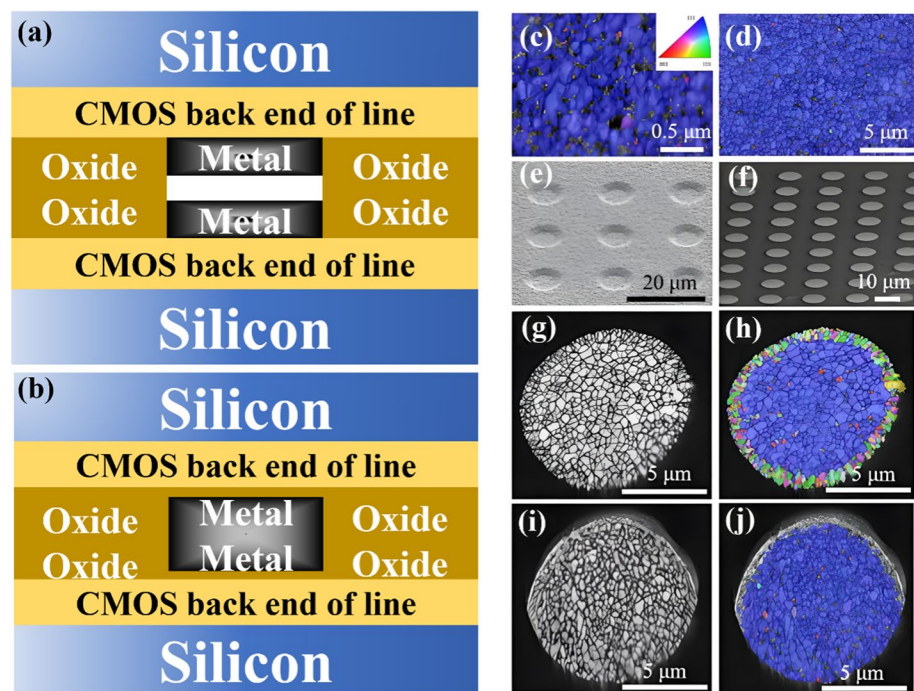
although capable of achieving low-temperature bonding in a short time, demands high surface flatness of the wafer and is complicated by the high hardness and rapid oxidation rate of copper, which increase the bonding difficulty.

4.2 Low-temperature Cu/SiO₂ hybrid bonding technology

For fine-pitch applications, bump-free Cu/SiO₂ interconnects are the most promising HB structures, achievable through DBI or hot-pressed bonding (TCB) techniques. This technology was developed by Ziptronix and is suitable for scalable HB processes [49]. The results show that DBI can reduce the interfacial parasitic capacitance by 80% in the spacing range below 20 µm compared with the traditional bumping process. It has been further demonstrated that the interconnect pitch can be scaled to less than 10 µm [70, 71], and the low-temperature HB process consists of two key steps, as shown in Fig. 9a, b. According to the analysis of the physical mechanism above, the direct bonding between SiO₂ at room temperature can be realized by hydrophilic modification of the surface. The copper–inter–copper bonding is then completed through a batch annealing method at temperatures ranging from 200 °C to 400 °C. The annealing heating rate should be limited to less than 5 °C/min to avoid microcrack defects at the edge of the copper pillar. Although this method has high production efficiency, the interface quality of SiO₂ is easily affected by both residual moisture and copper thermal expansion stress during annealing [72].

Additionally, another nanotwin Cu structure for potential Cu/SiO₂ HB utilizes the excellent diffusion traits of (111) crystalline copper, and the nanotwin copper film can be effectively bonded after 1 h of treatment at 200 °C. Based on this property, Chiu et al. [73] developed a composite structure of nanotwin copper and silica for low-temperature HB systems. Three morphologies were prepared and analyzed: nanotwin copper full-coverage films, and disc-shaped and pad-like hybrid structures (see Fig. 9c–j). SAT and scanning electron microscopy (SEM) characterization confirmed that the nanotwin copper film with 99% (111) preferential orientation can complete high-quality bonding at 250 °C.

Fig. 9 Illustrative diagram of the DBI technology's dual-step process. **a** Oxide bonding at room temperature. **b** Metal interconnect annealing; electron backscatter diffraction (EBSD) orientation image maps of the **c** Cu seed layer and **d** nt-Cu blanket film. **e** SEM image of dish-shaped nt-Cu fabricated by conformal plating and **f** pad-shaped nt-Cu post-CMP. **g** SEM and **h** EBSD images of the dish-shaped nt-Cu surface. **i** SEM and **j** EBSD images of the pad-shaped nt-Cu surface [70, 73]



The orientation ratio of copper grains (111) in the disc and pad mixed structure was 75.1% and 99%, respectively. The results show that the pad-like composite structure shows significant advantages in 250 °C bonding, but the synergistic bonding mechanism of nanotwin copper and silicon dioxide still needs to be further verified.

4.3 Wafer-level structure GaN/Si heterogeneous integration

Ren et al. [74] innovatively combined high-voltage gallium nitride (GaN) high electron mobility transistor (HEMT) devices and low-voltage silicon metal–oxide–semiconductor field-effect transistors (MOSFETs) to form a co-source and co-gate topology, and used device-level co-optimization to significantly suppress the parasitic effects of interconnect. However, this architecture requires a redesign of the process integration, such as ion implantation and annealing steps in the CMOS source-drain region after the completion of the GaN epitaxy process, so as to reduce the effect of high-temperature heat treatment on the dopant diffusion depth and concentration gradient. Limited by the integration complexity and density constraints of GaN devices, and the fact that HEMT devices can only realize N-type transistor functions, the device type is simplified, and it is difficult to meet the design requirements of ICs. On the other hand, the compatibility of high-voltage and low-voltage components in the power system needs to be solved urgently.

In response to the above challenges, Du et al. [75] took the lead in developing a 20 V GaN/Si CMOS 1P2M

monolithic heterogeneous integration technology based on a 6-inch (111) silicon substrate, and systematically completed process development and model construction. In this study, the growth of high-quality GaN heterostructures is achieved through the graphical selective epitaxy technology, the problem of GaN/Si-based P-well CMOS process matching is innovatively solved, and the platform-based device architecture is constructed, which breaks through the multi-level technical bottleneck in the integration of large-size silicon-based CMOS and GaN chips, and successfully develops core components such as high-performance bulk silicon PN diodes, bipolar transistors (PNP/NPN), and AlGaN/GaN HEMT. By solving the key problems such as resistance and capacity (RC)

parasitic effect and stress coupling mechanism, a new path is opened up for the design of gallium nitride ICs. The research team also built a heterogeneous integrated technology verification platform that integrates silicon-based CMOS and gallium nitride processes.

Ren et al. [76] discussed the use of integrated GaN and Si interconnects to achieve heterogeneous power distribution in large-scale chiplet-based systems. By employing a GaN-Si interconnect fabric with sub-10 μm bond pitch, Fig. 10a comprehensively demonstrates the application of the integrated GaN/Si interconnect fabric in a large-chip system and its heterogeneous power delivery capability at less than 10 μm bond pitch through a clear layout and detailed annotation. The system enables efficient power management and heat dispersion to support HPC and high-density integration. Figure 10b also shows the 3D

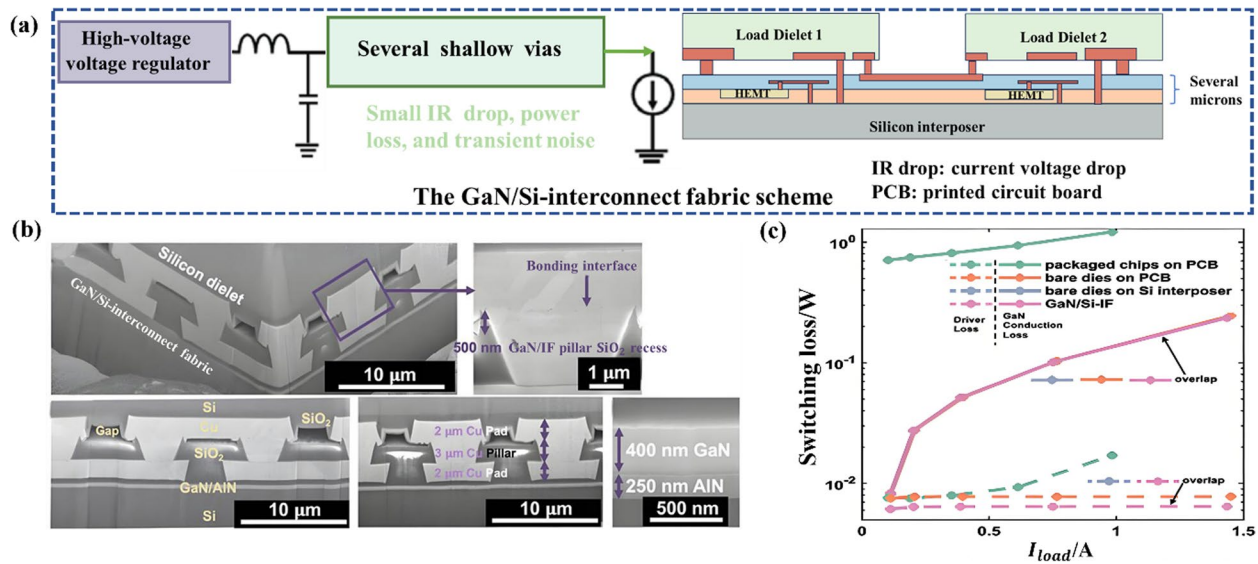


Fig. 10 **a** Illustration of the heterogeneous structure of GaN/Si CMOS. **b** 3D EM image of GaN/Si CMOS. **c** Driver and GaN losses vs. load current; efficiency can be improved by using silicon interposer and Cu–Cu bonding

EM image of GaN/Si CMOS. Figure 10c visually demonstrates the performance advantages of the co-source co-grid structure in a heterogeneous integrated system by comparing the gain with frequency, which makes full use of the high electron mobility of GaN materials and the mature process of Si CMOS technology to realize the complementary advantages of heterogeneous materials, and provides strong support for high-performance, low-power RF front-end systems. Through detailed data analysis, the paper demonstrates the significant advantages of integrated GaN/Si interconnect structures in terms of power distribution efficiency, thermal management, and signal integrity. Experimental results show that the system using this technology can improve the power distribution efficiency by more than 20% while reducing the thermal resistance by 30% compared with the traditional method. In addition, the data analysis revealed the critical impact of bond pitch on system performance, pointing out that sub-10 μm bond pitch was a key factor in achieving these improvements. These data not only verify the accuracy of the theoretical model, but also provide an important reference for subsequent system design and optimization.

4.4 Other integration technologies

At present, a three-dimensional multilayer bonding technology using Cu–Cu hybrid wafer bonding has also been developed. By applying robust Cu–Cu bonding in conjunction with the TSV process, a wafer-level multi-stacking structure was successfully demonstrated. This achievement

relies on a comprehensive understanding of the bonding interface, TSV, and edge engineering to avoid defects and ensure high yields. Chemical mechanical polishing (CMP) and surface treatment significantly impact the Cu/dielectric HB surface, particularly the planarization of the wafer back, which is a critical technology for subsequent robust bonding. Moreover, improvements in etching during TSV formation and considerations for integrated edge processing have optimized engineering aspects, enhancing bonding quality and reproducibility, leading to a high yield of electrical connections, including those at the wafer edges. This technology can be applied in various fields, including sensors, memory devices, logic devices, and their combinations.

5 Breakthroughs in hybrid bonding technology

In Cu–Cu HB manufacturing, the control of the bonding environment is one of the core challenges, and the key to this is to achieve a flat and contamination-free dielectric surface through CMP [77]. In order to reduce the risk of crystal shape and warpage during the bonding process, it is important to develop dielectric materials that are resistant to short-time annealing at low temperatures, as they need to be compatible with lower annealing temperatures and shorter process cycles. In addition, optimization of the ECD process can improve the overall bonding efficiency by adjusting the uniformity of the copper filling (e.g., inhibiting the formation of microvoids) and reducing the thermal budget required for annealing.

Table 2 CMP process parameters [79]

No.	CMP step	Process 1	Process 2	Process 3
1	Bulk planarization	High-pressure polish (HPP)	HPP	HPP
2	Barrier polish	HPP	HPP	Low-pressure polish (LPP)
3	Barrier over polish (by fixed time)	HPP (50 s)	LPP (50 s)	LPP (10 s)

CMP chemical mechanical polishing

In Cu–Cu HB, achieving flat and clean dielectric surfaces is a crucial step in the manufacturing process. The optimization of CMP technology is essential to meet this requirement [78]. CMP not only removes surface roughness but also achieves sub-nanometer surface planarity, which is vital for high-quality bonding. Research has shown that developing a three-step CMP process can enhance planarization accuracy, thereby improving the bonding interface quality. For example, Roy et al. [79] have demonstrated that developing a three-step CMP process can enhance planarization accuracy, thereby improving the bonding interface quality. In the study, a three-step CMP process was employed, which optimized various CMP conditions to achieve precise control over Cu dishing and surface roughness. This process combined high-pressure polishing (HPP) and low-pressure

polishing (LPP), and by precisely controlling the polishing time, significantly reduced the dishing of Cu pads. Table 2 presents the detailed information of the three process recipes, namely recipe 1, recipe 2, and recipe 3, each featuring distinct applied pressures and fixed durations in their respective sub-steps. Figure 11a illustrates the effect of CMP process parameters on the removal characteristics of bonded pad materials by analyzing the Cu removal rate and etch rate (ER) values at different polishing slurry flow rates and pad pressures. This is critical for optimizing the CMP process to achieve high-precision Cu bonding pad and barrier layer patterning. By adjusting these parameters, the removal rate of the barrier layer can be effectively controlled while ensuring the Cu removal efficiency, thereby improving the reliability and yield of the wafer-to-wafer bond. Figure 11b, c shows that the wafer is consistent between the center and edge of the beam using an optimized process 3. Bonded wafers perform well, with more than 90% of the bonded area and no visible voids. This study provides valuable reference and guidance for the development of more advanced CMP processes. In the future, finer Cu bonding pads and smaller-pitch CMP processes can be further explored to meet the needs of semiconductor device manufacturing with higher integration.

For example, in the study by Lee et al. [77], through the optimization of the three-step CMP process, significant improvements in surface planarity and electrical performance revealed by TSV were achieved, while overall

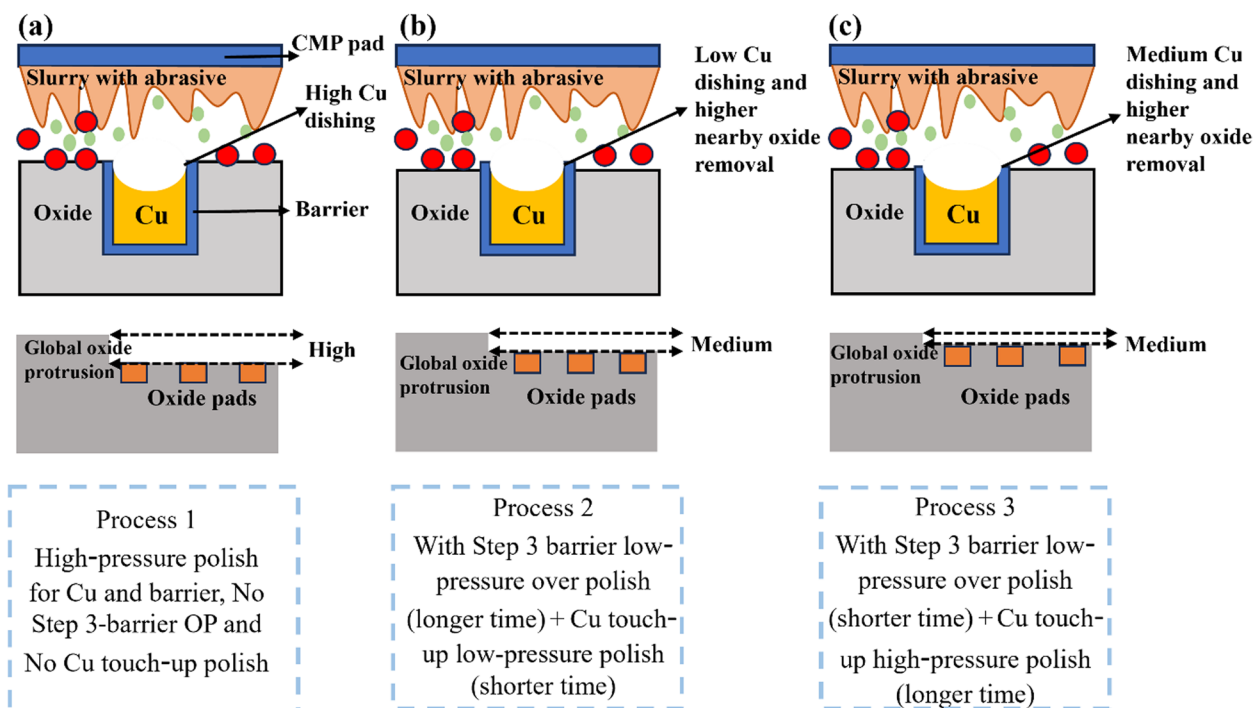
**Fig. 11** Schematic illustration of the fine-pitch Cu bond pad device under different conditions of CMP. **a** Process 1. **b** Process 2. **c** Process 3 [79]

Table 3 Results of electrical testing on design A samples annealed at 200 °C for an hour [78]

Build	Sample ID	Annealing temperature (°C)	#Die pass/# Die no defect	% Yield
1	#1-1	200	24/26	92
2	#2-1	200	20/20	100
3	#3-1	200	19/19	100
3	#3-2	200	13/13	100
3	#3-3	200	15/15	100

yield and reliability were enhanced. A significant technological advancement is the development of dielectric materials that endure lower annealing temperatures and shorter times to reduce the risk of wafer deformation and warpage during bonding. Traditional Cu–Cu bonding requires relatively high annealing temperatures (typically around 300 °C), which can lead to wafer warpage and stress concentration [78]. By optimizing dielectric materials and bonding processes, researchers have successfully reduced the annealing temperature to 150–200 °C while also shortening the annealing time. This low-temperature bonding technology not only reduces thermal stress but also enhances bonding efficiency. After conducting an annealing experiment at 200 °C for 2 hours, researchers shortened the annealing time to 1 hour for several samples and conducted long-term testing at 200 °C over several months. These samples were sourced from multiple different assembly batches. As shown in Table 3, out of the five tested samples, four achieved a perfect electrical test yield on void-free chip-to-wafer pairs, while one sample group reached a 92% yield. This result indicates that the experiment of shortening the annealing time has achieved significant success.

6 Current application domains

As semiconductor devices continue to evolve to higher density, lower power consumption, and heterogeneous integration, advanced packaging technologies (such as 3D ICs and chiplet architectures) have become one of the core paths to break through the bottleneck of Moore's law. Under this trend, Cu–Cu HB has become a key technology for high-bandwidth and low-latency chip system integration due to its ultra-high interconnect density ($<1 \mu\text{m}$ pitch), excellent electrothermal performance, and heterogeneous material compatibility. Especially in the fields of 5G communications, HPC, and AI chips, Cu–Cu HB significantly reduces parasitic resistance ($<10 \text{ m}\Omega$) and thermal resistance by replacing traditional solder bumps with direct copper interconnects, providing physical support for the co-optimization

of heterogeneous chips such as logic cells, memories, and optoelectronic devices.

6.1 Application fields

6.1.1 High-bandwidth memory (HBM)

Cu–Cu hybrid bonding technology is actively developed and adopted in HBM applications. By stacking multiple memory chips and bonding them through Cu–Cu HB, high-bandwidth and low-latency memory access is achieved.

6.1.2 3D chip integration

Chips with different functions, such as processors, caches, and logic chips, are vertically stacked and interconnected through Cu–Cu HB. This allows for more functional integration within limited space, enhancing the overall system performance and energy efficiency.

6.1.3 CMOS image sensors

Cu–Cu HB technology is used to bond pixel array chips with signal-processing chips, resulting in a more compact sensor structure. This improves image quality and processing speed while facilitating the miniaturization of camera modules.

Sony was the first to apply HB technology to contact image sensor (CIS), addressing the challenges of high-density pixels and electrical signal processing. In 2016, Sony launched the IMX260 image sensor based on Cu/SiO₂ HB technology [80], which was used in the Samsung Galaxy S7. The process included wafer activation, pre-bonding, two annealing steps, and high-temperature bonding [81], achieving excellent electrical reliability and image performance. In the past 2 years, Sony has reduced the interconnect pitch to below $2 \mu\text{m}$ [82], driving the miniaturization and high-performance of CIS image sensors. Figure 12 shows Sony's BI-SIC HB process flow: (i) TSV-free 3D CIS and processor IC integration (HB); (ii) SEM image of a 3D CIS mixed bonded to a processor IC; (iii) three-chip stacked (HB) structure. In 2018, based on wafer-level Cu/SiO₂ HB technology, companies such as Teledyne in the United States achieved heterogeneous integration of 250 nm InP heterojunction bipolar transistor (HBT) and 130 nm Si CMOS [83], demonstrating high-performance millimeter-wave ICs. The InP HBT wafer was processed with an Au/benzocyclobutene (BCB) interconnect layer and prepared with a SiO₂ surface embedded with micro-copper blocks. The Si CMOS wafer featured six layers of copper interconnect wiring. After bonding, the CMOS wafer substrate was thinned to $10 \mu\text{m}$ and electrically connected through Al pads. Tests showed that the HB process did not affect the performance of the transistors.

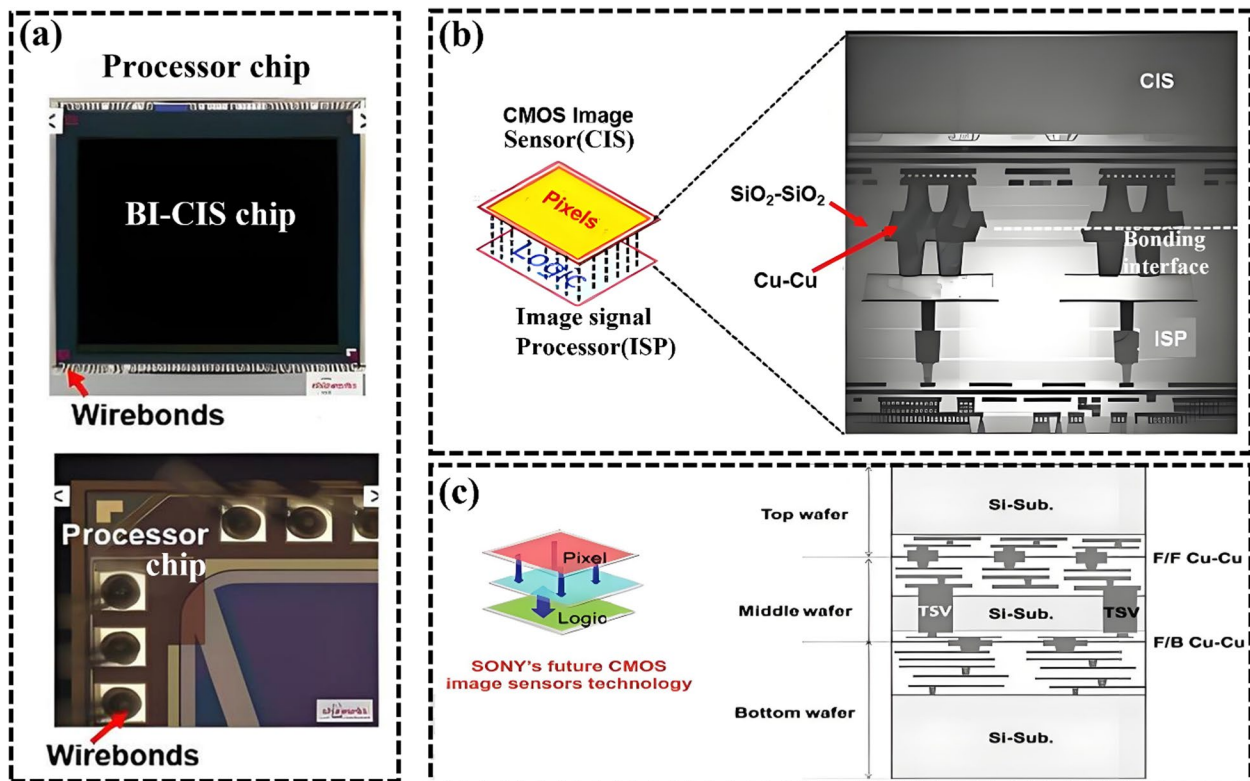


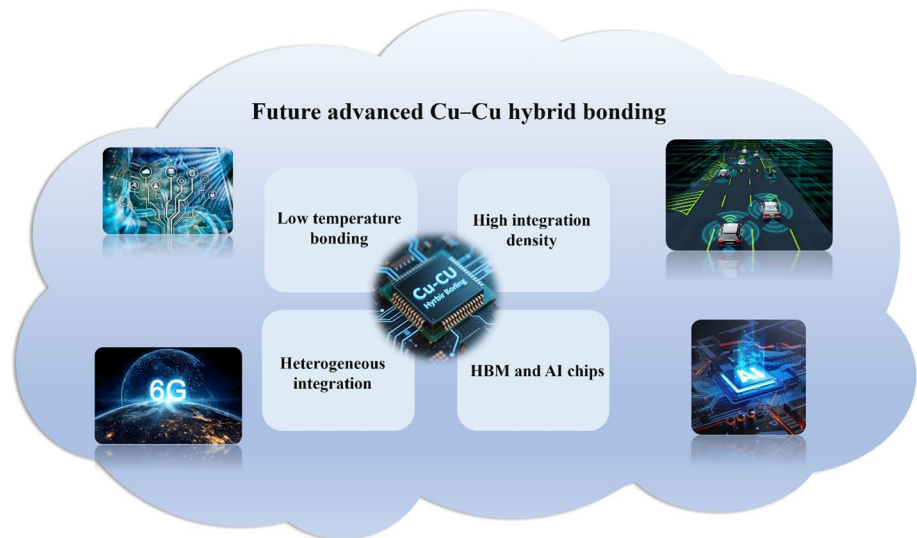
Fig. 12 Sony's CMOS image sensor manufactured by hybrid bonding. **a** TSV-free 3D CIS and processor IC integration (hybrid bonding). **b** SEM image of a 3D CIS mixed bonded to a processor integrated circuit. **c** Three-chip stacked (hybrid bonding) structure

Japanese companies such as Toshiba and Mitsubishi have been actively working on improving the interfacial reliability of HB technology by introducing novel auxiliary materials. In 2018, Toshiba developed a HB process based on an ultra-thin manganese (Mn) film [84]. This process begins with the deposition of a 300-nm-thick silicon dioxide (SiO_2) layer on the substrate using plasma-enhanced chemical vapor deposition (PECVD), followed by the deposition of a 2-nm-thick Mn film via ion plating. After activation treatment with nitrogen (N_2) plasma, the bonding process is carried out at temperatures ranging from 300 °C to 400 °C and a pressure of 2 MPa for 2 hours. Tensile test results indicate that at temperatures of 350 °C and above, the Mn–Mn interface exhibits sufficient mechanical strength to effectively suppress the diffusion behavior between copper (Cu) and SiO_2 , thereby significantly reducing the risk of electromigration. In 2021, Showa Denko Materials Co., Ltd [85] in Japan developed a polymer-to-polymer direct bonding technology based on photosensitive polymers. The process involved spin-coating a 5–10 μm thick layer of polyimide (PI) on a 200 mm silicon wafer, followed by pre-baking, ultraviolet (UV) exposure (300–600 mJ/cm^2), and developing in cyclopentanone. The PI was then cured for 2 hours each at 200 °C and 375 °C, and the surface roughness was reduced by CMP. Finally, the

wafers were bonded at 350 °C for 1 hour. SEM analysis of the bonded interface revealed no gaps or voids, indicating high-quality bonding. In 2022, the company completed the design and experimental validation of Cu/PI HB samples [86]. The design of Cu protrusions took into account the higher CTE of PI relative to Cu. The surfaces underwent treatment with a 5% citric or ascorbic acid solution, followed by a rinse with deionized water and drying with N_2 , before bonding. The bonding took place at 300 °C and 0.8 MPa in a nitrogen/hydrogen environment for 60 min. Focused ion beam (FIB)-SEM observations showed a high-quality bonding interface, with tests indicating optimal bonding quality at a bonding pressure of 5 MPa.

Samsung and SK Hynix are actively promoting the application of Cu–Cu hybrid bonding technology in HBM, aiming to achieve higher stacking density, reduce power consumption, and enhance performance. Samsung's HBM3 Icebolt features 12 layers of stacked 10 nm class 16 Gb dynamic random access memory (DRAM) chips with a capacity of up to 24 GB and a data transfer rate of 6.4 Gbps, achieving an astonishing bandwidth of 819 GB/s. Compared to traditional TSV technology, HB technology significantly reduces power consumption and enhances energy efficiency, leading to a 35% reduction in power consumption. This technology

Fig. 13 Cu–Cu hybrid bonding technology will facilitate integrated technology in the future, advancing AI chips and high-bandwidth memory



effectively lowers signal transmission losses through direct copper-to-copper connections and has also shown significant improvements in thermal management.

Current HB technology demonstrates significant advantages in improving chip integration density, enhancing data transmission bandwidth, and optimizing thermal management, particularly showing promising application prospects in fields such as HBM, 3D chip integration, and CMOS image sensors. However, several technical bottlenecks remain, including interface reliability, process complexity, material compatibility, and cost control for large-scale production. Although progress has been made through the introduction of novel auxiliary materials and optimization of process parameters (such as surface treatment and temperature control), further breakthroughs are still needed to achieve higher precision and lower-cost mass production.

7 Conclusions

This review centers on the core theme of “Cu–Cu HB technology,” systematically constructing a comprehensive framework spanning from physical mechanisms to macroscopic applications. It aims to provide a thorough overview of the research progress, critical technological breakthroughs, and future potential of this technology. Adopting a progressive logic of “theory–process–application–trend,” the review establishes a panoramic perspective from microscopic mechanisms to macroscopic industrial applications, emphasizing both technical depth and practical implementation. This approach offers a clear roadmap for subsequent research, presenting a holistic view of Cu–Cu HB technology, delivering a systematic summary of its current advancements while charting directions for future breakthroughs. As an advanced interconnect technology, Cu–Cu HB demonstrates

significant advantages in HPC, high-speed communication, and advanced packaging. By enabling direct copper-to-copper bonding, it achieves sub-micron or even nanoscale interconnect pitches, substantially enhancing chip integration density and data transmission bandwidth while reducing signal latency and energy loss. Furthermore, HB drives the development of 2.5D and 3D packaging technologies, facilitating vertical chip stacking to improve system performance and energy efficiency. As shown in Fig. 13, in future Cu–Cu mixed bonding will develop toward low-temperature bonding and high integration, while being extensively applied in heterogeneous integration and high-bandwidth storage and AI chips to meet the demands for high density and low latency. In summary, Cu–Cu HB technology is poised to continue driving electronic devices toward higher performance, integration density, and reliability, solidifying its role as a cornerstone of next-generation microelectronics.

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Data availability Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate This manuscript does not involve human or animal research.

Competing interests The authors have no conflicts of interest to declare.

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